1. Name your Solution (10 words max)

Secure AI Policy Compliance & Gap Analysis System

2. Describe in Brief Your Solution (100 words max)

Our solution is a privacy-first automated compliance platform powered by the **Mistral 7B Instruct v0.3 Large Language Model (LLM),** running entirely offline within your secure infrastructure. Utilizing a Retrieval-Augmented Generation (RAG) architecture, the system ingests policy documents and semantically maps them against NIST SP 800-53 controls using the quantized Mistral 7B model for high-precision, local inference. This approach ensures zero data leakage while automating gap detection, generating implementation roadmaps, and providing intelligent policy revisions, transforming manual audits into a streamlined, secure workflow.

3. USP of the Solution (100 words max)

Our solution features a Dual-Layer Cognitive Engine running entirely offline. First, an Intelligent Domain Partitioner (powered by **Mistral-7B-Instruct-v0.3-Q5_K_M**) autonomously dissects unstructured policy documents into semantic security domains (e.g., ISMS, Risk Management) without manual tagging. Second, it employs a RAG-based Semantic Mapper using all-MiniLM-L6-v2 embeddings to align these domains with NIST SP 800-53 controls at a granular level. This architecture allows for precise, context-aware gap detection and roadmap generation that outperforms generic keyword matching, all while ensuring zero data leakage by keeping sensitive policy data local.

4. Innovative Features (Bullet Points & 100 words max)

- Offline Data Sovereignty: Uses local Mistral-7B inference to process sensitive data without internet access.
- Semantic Control Mapping: Leverages 384-dimensional **all-MiniLM-L6-v2** embeddings for precise, context-aware NIST 800-53 control matching beyond simple keywords.
- Auto-Partitioning Engine: Dynamically segments unstructured PDFs into hierarchical domains (e.g., ISMS, Risk) without manual tagging.
- RAG-Driven Remediation: Retrieves granular NIST controls from a local vector store to generate specific, actionable policy revisions and compliance roadmaps.

5. Steps taken to complete the problem (Bullet Points & 500 words max)

Established Local Infrastructure:

- Set up a Next.js frontend and Flask backend to create a secure, air-gapped environment.
- Downloaded and deployed the Mistral-7B-Instruct-v0.3 LLM locally to enable offline inference.
- Initialized a local ChromaDB vector store to index the full NIST SP 800-53 control framework without external API calls.

Built the Document Ingestion Pipeline:

- Implemented a PDF processor using **PyPDF2** (with OCR fallback) to extract raw text from uploaded policy documents.
- Developed a custom "Chunk-and-Infer" mechanism where the local LLM autonomously segments unstructured text into logical security domains (e.g., ISMS, Risk Management).

Enabled Semantic Mapping:

- Integrated the all-MiniLM-L6-v2 embedding model to convert policy text into 384-dimensional vector embeddings.
- Mapped these embeddings against the local NIST database to identify relevant regulatory controls based on semantic meaning, not just keywords.

Executed Gap Analysis:

- Prompted the local Mistral-7B model to compare the organization's policy segments against the retrieved NIST standards.
- Systematically identified compliance gaps, classifying them by severity and mapping them to specific control failures.

Delivered Automated Solutions:

- Engineered a RAG workflow to auto-generate compliant policy revisions and actionable implementation roadmaps for each identified gap.
- Visualized the results in a modern dashboard, providing users with clear scores, detailed findings, and exportable reports.

6. Any challenges faced during Solution Development (250 words max)

- Optimizing Local Inference Latency: Running the 7B parameter Mistral model locally introduced significant latency.
- Handling Context Window Constraints: NIST SP 800-53 is distinctively extensive. Fitting relevant controls alongside detailed policy text into the LLM's context window required developing a dynamic chunking strategy that retrieves only the top-k most semantically relevant NIST chunks per query.
- PDF Structure Extraction: Parsing diverse organizational policy documents with complex layouts (headers, footers, multi-column text) proved difficult

- Minimizing Hallucinations: Ensuring the LLM didn't "invent" regulatory requirements was critical.
- Ambiguous Domain Classification: Automatically categorizing unstructured text into specific security domains.

7. Significant Impact Your Solution Can Make (150 words max)

Our solution revolutionizes regulatory compliance by automating **90%** of the manual audit workload, reducing the time to analyze a 100-page policy from weeks to minutes. By running entirely offline, it empowers highly regulated industries (healthcare, defense, finance) to adopt AI-driven compliance without risking data sovereignty or violating clearance protocols. The semantic accuracy of vector-based NIST mapping drastically minimizes false positives compared to keyword searches, ensuring organizations focus resources only on genuine gaps. Furthermore, the auto-generated implementation roadmaps transform static audit findings into dynamic action plans, significantly accelerating the remediation cycle and improving overall organizational security posture.

8. Drive link (where we can find codes and supplementary material with proper readme to validate Your Solution)

https://drive.google.com/drive/folders/1-2gqqgc8YvDwQj0PaquZLmW5Y272vCQt?usp=sharing